

ERES-Pellis Spectral Bridge: Center-Balance as a Non-Self-Adjoint Operator

10/10 USE MATH — Why, How, and Next-Actions, Built on the Balancing-Extremes Proof-of-Work

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Status: Hypothesis-tier working note — internal use. Not yet packaged for Dr. Stergios Pellis (University of Ioannina) pending Track B sequencing.

WHY

The Balancing-Extremes Proof-of-Work (DeepSeek derivation, Grok synthesis, and the public "Balancing Life's Extremes" article) already contains a **fully closed-form, numerically validated optimization model**. That model has a property its authors didn't push on: its stability structure is a **2×2 Jacobian operator**, and a small, honest generalization of one term turns that operator **non-self-adjoint** — which is exactly Pellis's native mathematical territory.

This matters for the Pellis collaboration for three concrete reasons:

1. **It's a self-contained, low-risk artifact.** It doesn't require Pellis to first absorb the full BERA spec, the UBIMIA manual, or PSCI-hat's derivation. It's a two-variable, fully worked toy model with closed-form eigenvalues — the kind of minimal example a spectral theorist can evaluate in minutes.
 2. **It's already "free" work.** No new claim is being made about BERA, PSCI-hat, or REAL — only a *proposed mapping*, clearly labeled as a hypothesis. This respects the existing sequencing: lead with free, self-contained contributions; hold the heavier Track B material until Pellis responds.
 3. **It demonstrates exceptional-point structure using ERES's own existing proof, not borrowed physics.** The "extremes collapse / center balances" result already proven in the Proof-of-Work is mathematically the same phenomenon (a discriminant crossing zero) that defines exceptional points in non-Hermitian/non-self-adjoint operator theory. That convergence wasn't designed — it falls out once leverage is allowed to be asymmetric, which is also a *more realistic* refinement of the existing model on its own terms.
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HOW — The Bridge Construction

1. Starting point: the existing closed-form model

From the Proof-of-Work (Simple Quadratic Model):

$$C(R, W) = \alpha R + \beta W + \gamma RW - \delta R^2 - \epsilon W^2$$

with stationary conditions $\partial C / \partial R = \partial C / \partial W = 0$ giving the symmetric Hessian:

$$H = \begin{pmatrix} -2\delta & \gamma \\ \gamma & -2\epsilon \end{pmatrix}$$

H is **symmetric** (self-adjoint) — its eigenvalues are guaranteed real:

$$\lambda_{\pm} = -(\delta + \epsilon) \pm \sqrt{(\epsilon - \delta)^2 + \gamma^2}$$

Negative-definiteness (comfort optimum, not saddle) requires $\gamma^2 < 4\delta\epsilon$ — exactly the discriminant condition already derived in the Proof-of-Work as the "interior optimum exists" guarantee.

2. The generalization: asymmetric leverage

The model assumes work's effect on family comfort equals family's effect on work comfort (γ appears once, symmetrically). That's a simplifying assumption, not a structural necessity. Split it:

$$\frac{\partial C}{\partial R} = \alpha + \gamma_1 W - 2\delta R, \quad \frac{\partial C}{\partial W} = \beta + \gamma_2 R - 2\epsilon W$$

where γ_1 = how strongly profit-effort raises the marginal value of relational investment, and γ_2 = how strongly relational investment raises the marginal value of profit-effort. These need not be equal — and in lived experience, usually aren't.

This produces the generalized Jacobian (the "**leverage operator**" J):

$$J = \begin{pmatrix} -2\delta & \gamma_1 \\ \gamma_2 & -2\epsilon \end{pmatrix}$$

J is **non-self-adjoint** whenever $\gamma_1 \neq \gamma_2$ ($J \neq J^T$). This is the bridge.

3. Spectrum of the leverage operator

$$\lambda_{\pm} = -(\delta + \epsilon) \pm \sqrt{(\epsilon - \delta)^2 + \gamma_1 \gamma_2}$$

Three regimes:

Condition	Regime	Interpretation
$\gamma_1 \gamma_2 > -(\epsilon - \delta)^2$	Real, distinct eigenvalues	Overdamped return to balance — comfort relaxes monotonically to the optimum
$\gamma_1 \gamma_2 = -(\epsilon - \delta)^2$	Exceptional point	Eigenvalues coalesce, J becomes defective (single eigenvector, geometric multiplicity 1)
$\gamma_1 \gamma_2 < -(\epsilon - \delta)^2$	Complex conjugate pair	Oscillatory approach to balance — comfort overshoots and corrects in cycles

The exceptional point is the precise mathematical event behind "extremes collapse, center balances" once leverage is allowed to be directional rather than mutual: it's the boundary where stable convergence to center-balance turns into oscillatory correction.

4. Left/right eigenvectors and biorthogonality

For non-symmetric J , right eigenvectors v_i ($Jv_i = \lambda_i v_i$) and left eigenvectors u_i ($u_i^\top J = \lambda_i u_i^\top$) are generally **not the same vectors** and are not orthogonal to each other — only biorthogonal across different eigenvalues: $u_i^\top v_j = 0$ for $i \neq j$.

As $\gamma_1 \gamma_2 \rightarrow -(\epsilon - \delta)^2$ (approaching the exceptional point), u_+ and v_+ become parallel, and the eigenvector condition number

$$\kappa(V) = \|V\| \cdot \|V^{-1}\|, \qquad V = [v_+ \mid v_-]$$

diverges. This divergence is a direct, computable **sensitivity measure** of how close the balance system sits to structural collapse — not a metaphor, a number.

5. Proposed mapping (hypothesis-tier — to be tested with Pellis, not asserted)

ERES construct	Candidate spectral analogue
BERA Disruption Index	Normalized departure-from-normality of J : $\ JJ^* - J^*J\ _F$, or the eigenvector condition number $\kappa(V)$ — both diverge at the exceptional point

ERES construct	Candidate spectral analogue
PSCI-hat	Biorthogonal overlap $u_+^T v_+ / (\ u_+\ \ v_+\)$ — a coherence measure that collapses toward zero precisely at the exceptional point
REAL formula ($\frac{E \cdot M \cdot R}{T \cdot S}$)	$\text{Re}(\lambda)$ as dissipation/decay rate (energy term), $\text{Im}(\lambda)$ as resonance frequency — both functions of the same $(\delta, \epsilon, \gamma_1, \gamma_2)$ parameter set

These are **explicitly proposed**, not established equivalences. The value of the bridge is that it gives a concrete, two-parameter, closed-form system on which any of these mappings can be tested numerically before being asserted in the BERA measurement spec.

6. Consistency check with the sigmoid bound

The Proof-of-Work's advanced (bounded) model wraps the raw score in a sigmoid: $C = \sigma(\Psi)$. At any stationary point, $\nabla \Psi = 0$, so:

$$\text{Hess}(C) = \sigma'(\Psi) \cdot \text{Hess}(\Psi)$$

Since $\sigma' > 0$ everywhere, the sign and eigenstructure of $\text{Hess}(C)$ at the critical point match $\text{Hess}(\Psi)$ exactly. **The exceptional-point analysis above survives the bounded model unchanged** — it's a property of the local Jacobian, not of which monotone bounding function sits on top. This means the bridge applies to the full 10/10 construct already in the corpus, not only the toy quadratic.

NEXT-ACTIONS

- Keep this internal for now.** Per established sequencing (lead with free contributions, hold Track B until Pellis responds), this stays as a working note — not yet sent.
- Numerically extend the existing code.** The Proof-of-Work already has a working Python implementation of the symmetric model. The asymmetric generalization needs only: replace the single `gamma` with `gamma_1`, `gamma_2`; compute `np.linalg.eig(J)` instead of the closed-form symmetric solution; sweep `gamma_1*gamma_2` across the exceptional-point boundary and plot eigenvalue coalescence. (I can build this artifact now if useful.)
- Decide on instrument status.** Either register this as a new, narrowly-scoped instrument (e.g., `ERES-COMFORT-SPECTRAL-2026-001`) or fold it as an appendix into the existing BERA measurement spec v0.2.1 — recommend the former, since it's self-contained and shouldn't load extra surface area onto the spec while Track A is still pending.

4. **If/when Pellis responds and Track B opens:** this becomes the candidate first worked example — small, closed-form, in his language, with an explicit "here is where I'd want your judgment" question (e.g., "is eigenvector condition number $\kappa(V)$ or the Henrici departure-from-normality the more defensible disruption metric for your operators?").
 5. **Do not assert the Disruption Index / PSCI-hat / REAL mappings as settled.** They're flagged hypothesis-tier here deliberately — any version sent outward should carry the same flag until Pellis (or independent derivation) confirms or revises them.
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